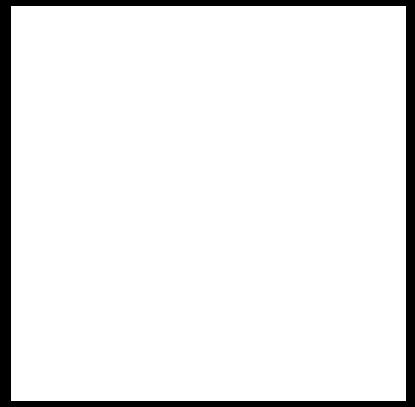
ual: creative computing institute

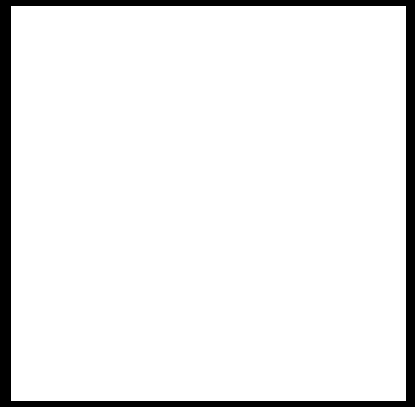
Week 1 - Creative Industry Data x Raw Data is an Oxymoron

Louis McCallum

Spring 2026



Housekeeping



Schedule

- 9.30-1.30 Tuesdays
- Attendance Mandatory
- Always in Person in Peckham (Greencoat)

Tutors

- Louis
- Loo-ee
- He/him
- Course Leader MSc Applied Machine Learning for Creatives
- Makes Music and Art
- Makes software for Music and Art

Tutors

- Zhou Zhou
- him/his — “jo-jo”
- Joint PhD Researcher: UAL and London Contemporary Dance School
- Research Focus: Co-creation between human and machine intelligence, using signal processing and AI in dance.
- Practice: Media Artist & Director exploring agency, ethics, and embodied interaction in performance



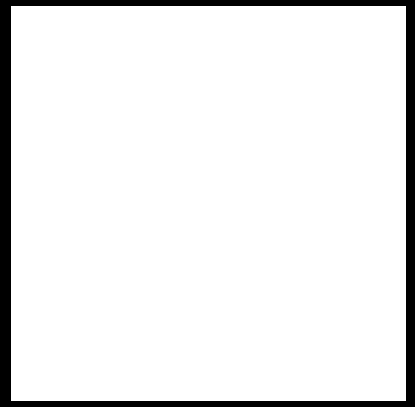
Assessment

- Submit in Week 10
- Portfolio of 3 Small Projects
- Open ended brief to pick 3 topics from the term and conduct an independent investigation using that approach for each topic
- Either expanding on given data with a new investigation, or exploring new data with provided techniques

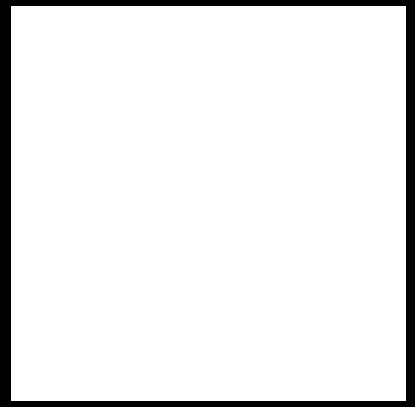
Creative Industries

Creative Industries

- Tends to be more of a government policy word than anything else
- “The place where art, business, and technology meet in economic activity”
  - Terras, M., Jones, V., Osborne, N. and Speed, C., 2024. Data-Driven Innovation in the Creative IndustriesTaylor & Francis.
- “at the crossroads of the arts, business and technology” (United Nations, 2008, p. iii)



Why do they use data?



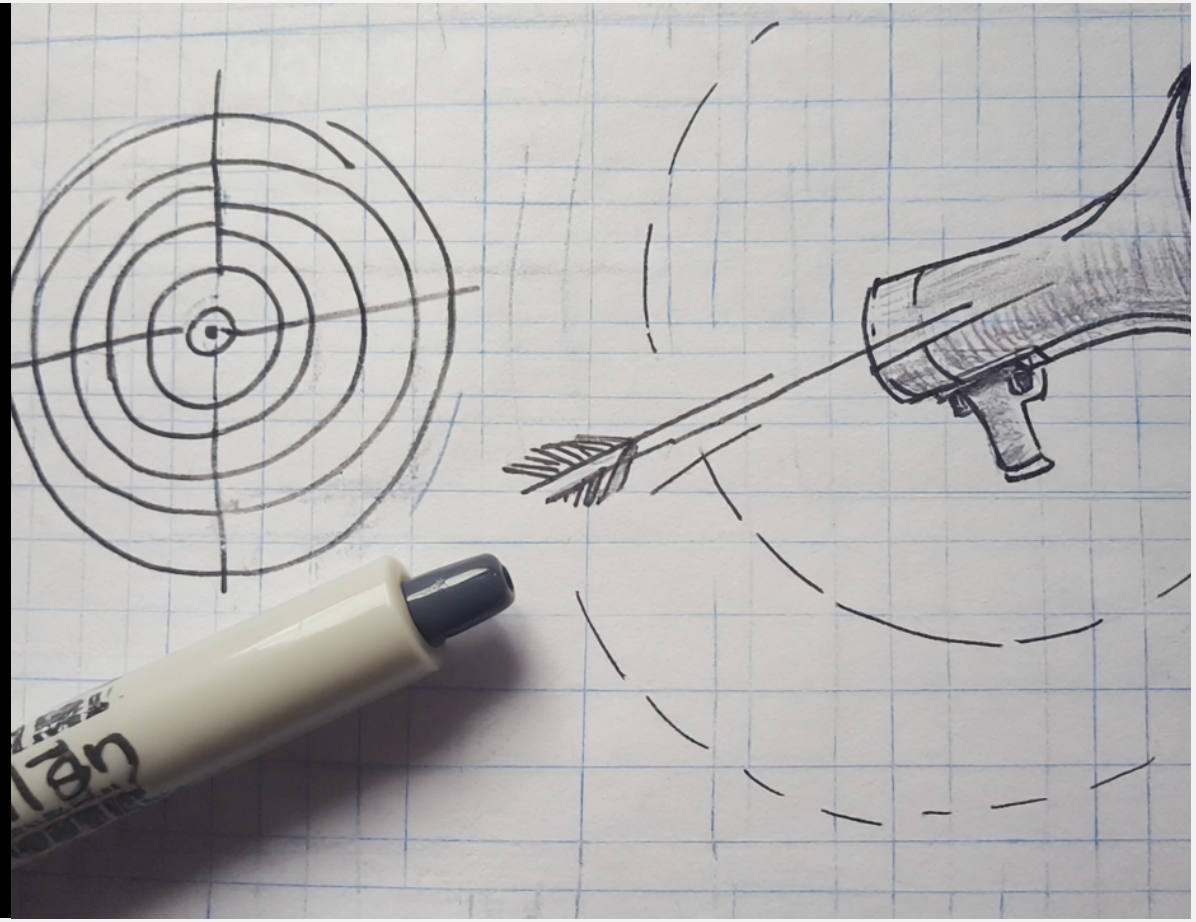
Data Science in the Creative Industries

How is this campaign/ release performing vs. expectations?



Data Science in the Creative Industries

Who is our audience and how do we reach them?



Data Science in the Creative Industries

Does release day significantly impact streaming performance?

When should we release this content for maximum impact?



Data Science in the Creative Industries

Which content will perform best?



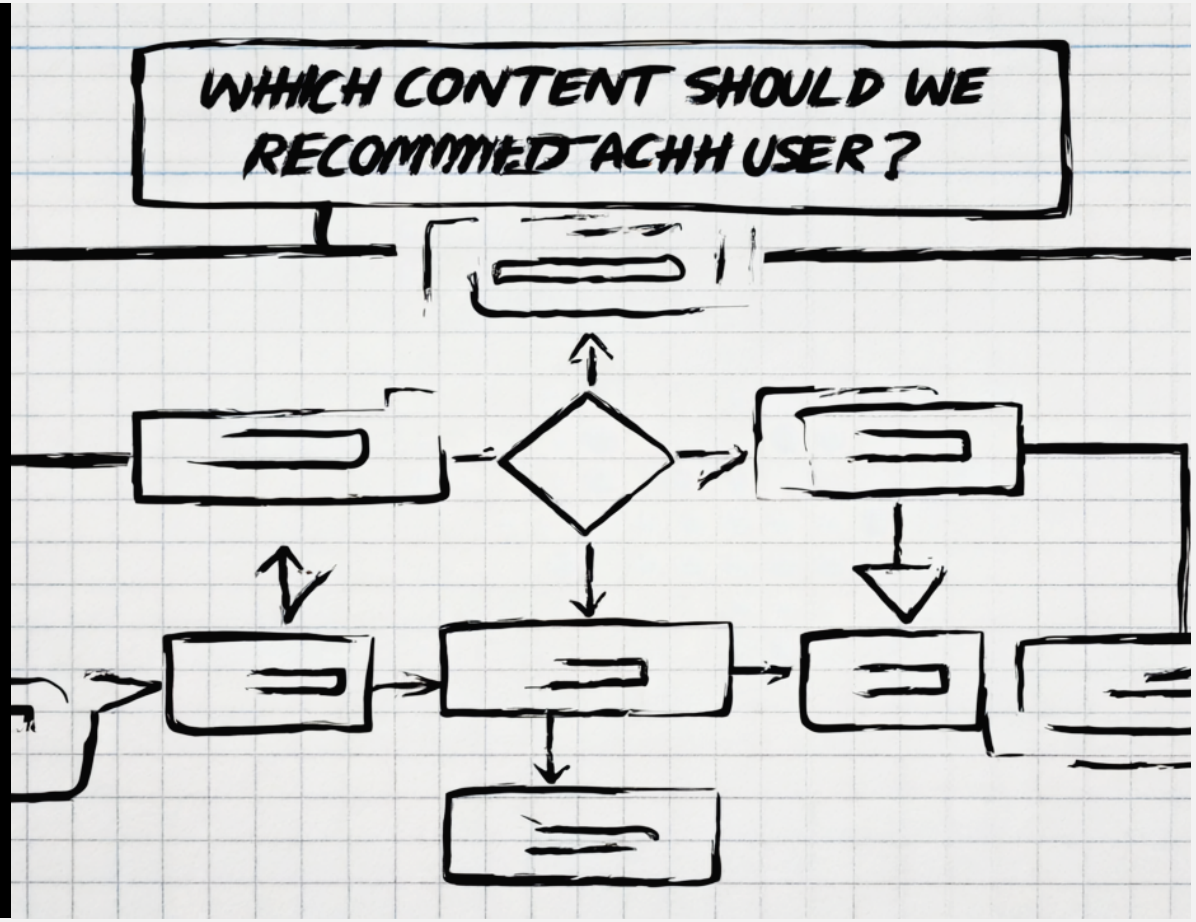
Data Science in the Creative Industries

We've just acquired a new content library, what are all the genres of the items?



Data Science in the Creative Industries

Which content should we recommend to each user?



Data Science in the Creative Industries

Is what we're doing ethical?



Data Science in the Creative Industries

Is what we're doing legal?



Stakeholders

- Musicians, singers, songwriters
  - streaming analytics, royalty calculations, playlist algorithms
- Visual artists, photographers
  - platform algorithms, sales data, exposure metrics
- Writers, authors, journalists
  - readership analytics, publishing algorithms, content recommendations
- Filmmakers, directors, producers
  - viewership data, distribution algorithms, audience insights
- Social media influencers
  - engagement metrics, algorithm changes, brand partnership data
- Podcasters
  - listener analytics, discovery algorithms, monetisation data
- Game developers
  - player behaviour data, recommendation systems, sales analytics

Stakeholders

- Streaming services (Spotify, Netflix, YouTube)
  - recommendation engines, content curation
- Social media platforms (Instagram, TikTok, Twitter)
  - content moderation, algorithm design
- E-commerce platforms (Amazon, Etsy)
  - search rankings, seller analytics
- Gaming platforms (Steam, mobile app stores)
  - discovery algorithms, review systems
- Publishing platforms (Medium, Substack)
  - content distribution, reader matching
- Music listeners
  - personalised recommendations, privacy concerns, listening habits tracked
- Readers
  - book recommendations, reading behaviour analysis, targeted advertising
- Viewers (film/TV/video)
  - viewing patterns, content suggestions, demographic profiling
- Art collectors & enthusiasts
  - purchase recommendations, market insights, authenticity verification

RecSys’24

- Is a key ACM Conference
- Has an industry track and this is often more interesting because companies
  - Have more data
  - Have live audiences to A/B test stuff
- Here are some interesting ones from 2024

RecSys'24

- "…fresh content from users is also critical for platform to build a high-quality supply in a sustainable way [8]. As such, it is critical for the recommendation system to find content that are both interesting for users to watch and **inspiring for users to generate their own content**"

Zhen Zhang, Qingyun Liu, Yuening Li, Sourabh Bansod, Mingyan Gao, Yaping Zhang, Zhe Zhao, Lichan Hong, Ed H. Chi, Shuchao Bi, and Liang Liu. 2024. Co-optimize Content Generation and Consumption in a Large Scale Video Recommendation System.



Co-optimize Content Generation and Consumption in a Large Scale Video Recommendation System

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RecSys'24

- Query Recommendations = incomplete searches / autofill
- "is the prefix query "new" related to a music intent or to a news podcast?)"

Henrik Lindstrom, Humberto Jesus Corona Pampin, Enrico Palumbo, and Alva Liu. 2024. Encouraging Exploration in Spotify Search through Query Recommendations.



Encouraging Exploration in Spotify Search through Query Recommendations

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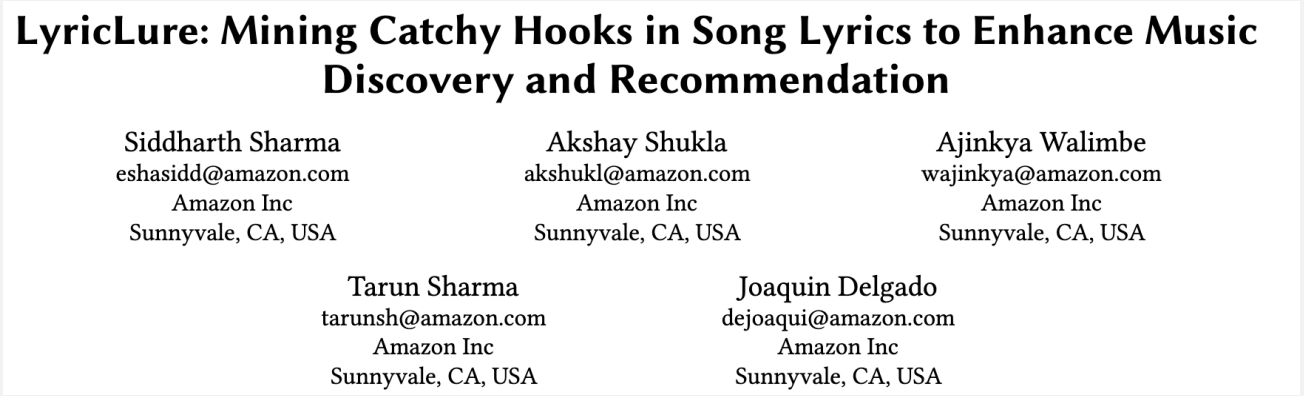
Enrico Palumbo
enricop@spotify.com
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Alva Liu
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Spotify
Stockholm, Sweden

RecSys'24

- Integrating lyrics into existing lexical search index poses difficulties due to lyrics text length.
- This paper introduces a proactive approach to extract catchy phrases from song lyrics

Siddharth Sharma, Akshay Shukla, Ajinkya Walimbe, Tarun Sharma, and Joaquin Delgado. 2024. LyricLure: Mining Catchy Hooks in Song Lyrics to Enhance Music Discovery and Recommendation.



LyricLure: Mining Catchy Hooks in Song Lyrics to Enhance Music Discovery and Recommendation

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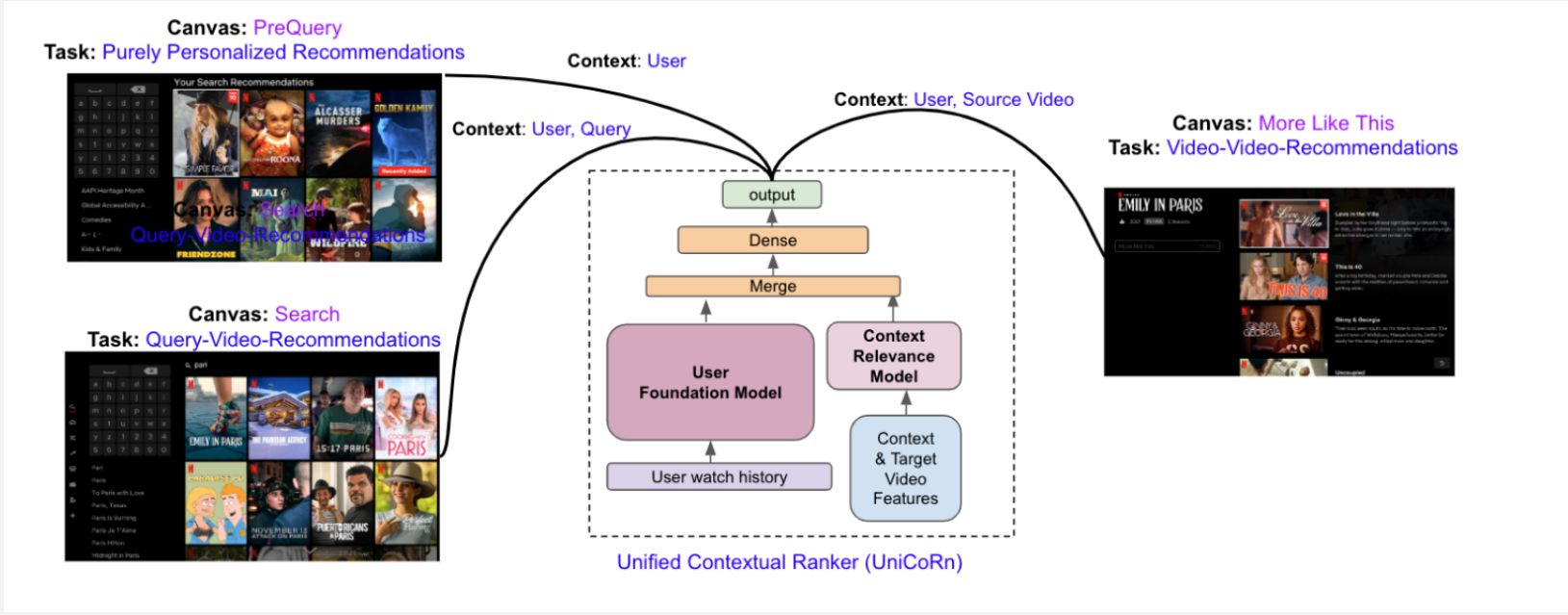
Tarun Sharma
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Sunnyvale, CA, USA

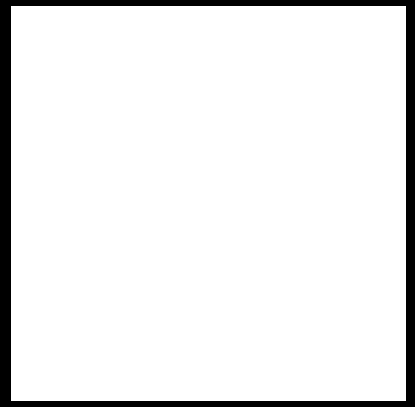
Joaquin Delgado
dejoaqui@amazon.com
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RecSys'24

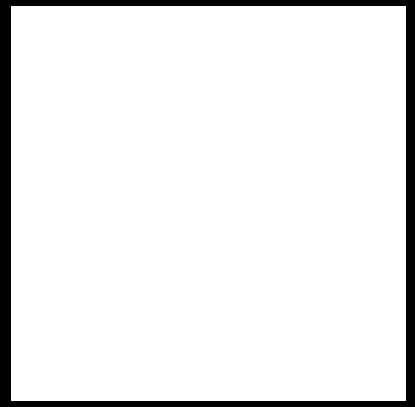
- Using the same model for different tasks can be **more efficient** to run and maintain

Moumita Bhattacharya, Vito Ostuni, and Sudarshan Lamkhede. 2024. Joint Modeling of Search and Recommendations Via an Unified Contextual Recommender (UniCoRn).





Data Science



Data Science

- Improving decision making through the analysis of data

In this unit

- Everything will be an **actionable insight** that is directly applicable to industry because I am a nice person who believes you deserve a job

In this unit

- But also at all times we will remember that all **data is of people**, and that **people** are on the receiving end of the actions of informed by data
- And that
  - exploitation of minorities
  - undervaluing of creatives
  - erasure of personal experience
- are all very real outcomes

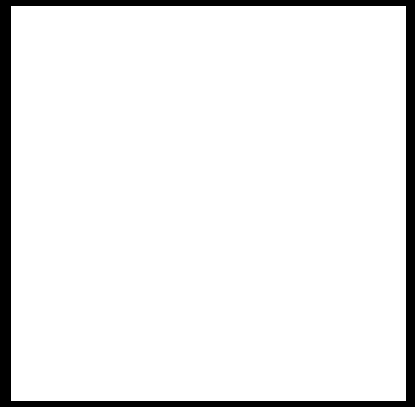
In this unit

- With great power comes great responsibility and if you want to be kissing Kirsten Dunst in the rain upside down, you need to tread carefully

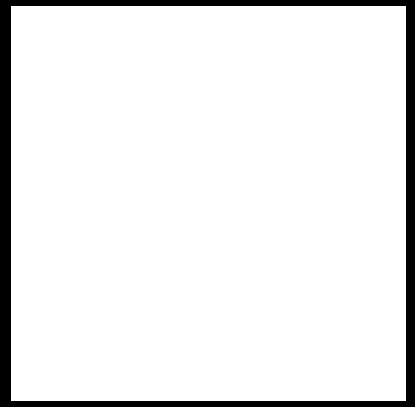


In this unit

- For every week we will address
  - What is the question I can answer?
  - What data will I need?
  - What stats / modelling do I do?
  - How do I interpret the results?
  - Can I express a confidence?



What’s in a dataset?



What is Data?

- Errrrrr…..its actually what are data.
- Data is technically plural, datum is singular

Where did it come from?

- “All data are cultural artefacts, created by people…at a time, in a place”
  - Yanni Loukissas, All data are local
- Digital humanities theorist Johanna Drucker says we should use *capta* (meaning taken) rather than *data* (meaning given)

Where did it come from?

- “Raw Data is an oxymoron”
  - Geoffrey Bowker and Lisa Gitelman
- All data is in some way curated (or cooked), collected for a reason, recorded in a certain way for a reason, encapsulating both prejudice and intent

Structured and Unstructured Data

- Structured data
  - Every item in the dataset has the same set of attributes
- Unstructured
  - Each item might have its own unique internal structure
  - e.g. a webpage, music file or image
- We can exact structured attributes from unstructured data!

Data Types - Quantitative

- Discrete (can’t be divided)
  - The number of songs on an album
  - The number of tickets sold for a play
  - The number of mentions of #LondonFashionWeek
  - Date of the album release
  - Daily Clicks on banner

Data Types - Quantitative

- Continuous (almost any numeric value)
  - The pitch of a recorded sound
  - Time taken to build a concert hall
  - Percentage increase in the auction price of a painting
  - Average weekly box office takings

Data Types - Qualitative

- Nominal or Categorical (no order)
  - Genre of film
  - Type of Subscription
  - Locations of Opera houses

Data Types - Qualitative

- Ordinal (intrinsic order, but not numeric, can’t do arithmetic)
  - Rankings of 8 channel noise compositions for prestigious Arts Prize™
  - Top 10 Films to Watch
  - Sometimes dates (when order but not interval is important)

Creative Industry Data

Types of Data

- User behaviour
  - Clicks, purchases, listens / views, likes, shares etc..
- Content
  - Structured metadata
  - Unstructured media data
- Performance metrics
  - streams, views, completion rates, revenue